

Benefits of Study Abroad and Working Memory on L2 Morphosyntactic Processing

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Introduction

Inflected languages mark semantic information with lexical cues (e.g., explicit subjects) and morphological cues (e.g., suffixes) (Evans, 2003), but adult second language (L2) learners have difficulty processing the latter. This difficulty is exacerbated in sentences containing both cues encoding the same meaning (Ellis, 2006), in learners with low proficiency levels (Hopp, 2010) or low working memory (WM) span (Sagarra & Herschensohn, 2010), and in native speakers of a morphologically poor or null language (Sagarra & Ellis, 2013). To address this problem, we investigate whether study abroad SA and high WM span can facilitate the processing of inflectional morphemes in low-proficiency adult learners whose first language (L1) is morphologically poor.

Our first independent variable, (SA), refers to temporarily living and studying the L2 in the L2 environment (Collentine & Freed, 2004). It is unclear whether the abundance of inflectional morphemes (input quantity), lack of redundant lexical cues (input quality), and increased communicative opportunities typical of SA contexts facilitate attention to L2 inflectional morphology. Some studies show that SA aids in the processing of L2 morphosyntactic agreement violations (Faretta-Stutenberg, 2014; Sagarra, 2017; Sagarra, Sanz, & Rodriguez, 2016, with behavioral data; Grey, Williams, & Rebuschat, 2015; LaBrozzi, 2009, 2012), whereas others report no SA effects (Faretta-Stutenberg, 2014). The investigation of SA is critical because more naturalistic studies have higher ecological validity than controlled classroom and lab experiments.

Our second independent variable, WM, consists of the activation and maintenance of short-lived memory items while performing sometimes complex and time-consuming cognitive tasks (Barrouillet & Camos, 2007). Studies examining WM and L2 morphosyntactic processing with non-study abroad (NSA) learners are inconclusive. Faretta-Stutenberg (2014), Havik, Roberts, van Hout, Schreuder, and Haverkort (2009), Keating (2010), Linck and Weiss (2011, 2015), Sagarra (2008, 2017), Sagarra and Herschensohn (2010), Sagarra et al. (2016), Sanz, Lin, Lado, Stafford, and Bowden (2016), and Serafini and Sanz (2016, with low-proficiency learners) all report WM effects. In contrast, Foote (2011), Coughlin and Tremblay (2013), Frost, Siegelman,

Narkiss, and Afek (2013), Grey, Cox, Serafini, and Sanz (2015), Sagarra (2000), and Serafini and Sanz (2016, with high-proficiency learners) report no WM effects. According to Sagarra (2017), this variability depends on L2 proficiency and the cognitive load of both the task and the WM test. WM effects with low-proficiency NSA learners are magnified with SA learners as they (i) have little time for planning and self-monitoring (so as to keep the conversation flowing) (Lafford, 2006) and (ii) have fewer attentional resources available to process the L2 because they are already using some to constantly suppress the L1 (Sunderman & Kroll, 2009; Tokowicz, Michael, & Kroll, 2004). Thus, scholars report WM effects in intermediate learners (LaBrozzi, 2009, 2012) but not in advanced learners (Grey et al., 2015) or in intermediate learners performing a taxing task (Faretta-Stutenberg, 2014) (Faretta-Stutenberg's timed grammaticality judgment task is cognitively more taxing than LaBrozzi's untimed reading task.) The examination of the role of learning context (NSA, SA), individual differences in WM, and L2 morphosyntactic development is essential to understanding how external and internal factors interact and affect SLA.

In this study, we examine the effects of SA and WM on low-intermediate proficiency-level Spanish learners' sensitivities to adjacent SV number agreement violations. In Spanish and English, verbs agree in person and number with their subjects, and verbs indicate SV agreement via verbal suffixes. However, Spanish is a pro-drop language with a rich morphological structure, whereas English requires an explicit subject and has poorer morphology (e.g., it lacks person markers in the past and only distinguishes between third-person singular and the rest of persons in the present). As a result, English speakers are more lexically biased than Spanish speakers.

To shed light on the role of SA and WM on L2 morphosyntactic processing, we employed eye-tracking methodology. We chose this technique because it measures early and late processing mechanisms (self-paced reading only targets late processing). In addition, it allows participants to read complete sentences at their own pace and make regressions.

Previous Literature

L2 Morphosyntactic Processing

According to Associative-Cognitive Theory, learning that a stimulus is related to a specific outcome makes the later learning of a new stimulus associated with the same outcome more difficult (Kruschke & Blair, 2000). The blocking of later-experienced cues by earlier ones guides attention to L2 cues in the short term (blocking) and long term (transfer), and modulates L2 ultimate attainment (Ellis, 2006). Adult learners focus on reliable cues before available cues (MacWhinney, Pléh, & Bates, 1985) and favor lexical cues when they face difficult structures (Ellis & Sagarra, 2011; Han & Liu, 2013) or have low L2 proficiency (e.g., Ellis & Sagarra, 2010; Parodi, Schwartz, & Clahsen, 2004; Steinhauer, White, & Drury, 2009). Relevant to our study, many studies show that higher-proficiency learners are sensitive to morphosyntactic violations (e.g., behavioral studies: Foote, 2011; Keating, 2010; Sagarra & Ellis, 2013; neurocognitive studies: Rossi, Gugler, Friederici, & Hahne, 2006), but lower proficiency ones are not (e.g., behavioral studies: Sagarra, 2008; Sagarra & Herschensohn, 2010; neurocognitive studies: Morgan-Short, Sanz, Steinhauer, & Ullman, 2010; Rossi et al., 2006).

WM Effects on L2 Morphosyntactic Processing

Studies on WM and L2 morphosyntactic processing are inconclusive. Thus, some scholars find WM effects (Faretta-Stutenberg, 2014; Havik et al., 2009; Keating, 2010; Sagarra, 2008, 2017; Sagarra & Herschensohn, 2010), whereas others report no WM effects (e.g., Foote, 2011; Coughlin & Tremblay, 2013). Longitudinal studies on WM and L2 grammar development are equally controversial, with some revealing WM effects (Linck & Weiss, 2011, 2015; Sagarra, 2017; Sanz et al., 2016; Serafini & Sanz, 2016, with lower, but not higher, proficiency learners) and others revealing no WM effects (Frost et al., 2013; Grey et al., 2015; Sagarra, 2000) (see Mitchell et al., 2015, for a review).

Sagarra (2017) explains these discrepancies in terms of L2 proficiency and the cognitive load of both the task and the WM test. Thus, studies using a taxing WM test (a test with a processing activity performed under time-pressure), and low-proficiency learners reveal WM effects on L2 grammar development (Havik et al., 2009; Linck & Weiss, 2011, 2015; Sanz et al., 2016; Serafini & Sanz, 2016) and L2 morphosyntactic processing (Faretta-Stutenberg, 2014; LaBrozzi, 2009). However, studies with a non-taxing WM test, and low- (Sagarra, 2000) or high-proficiency learners (Frost et al., 2013; Grey et al., 2015; Serafini & Sanz, 2016) show no WM effects on L2 grammar development. Finally, Sanz et al. (2016) did not find WM effects on L2 grammar development with a taxing WM test and a low proficiency level, likely because the task was too easy.

WM and L2 Morphosyntactic Processing in SA Contexts

Research on SA on SLA has produced mixed findings (see Sanz, 2014, and Yang, 2016, for reviews). Off-line studies on SA and L2 grammatical accuracy are also inconclusive, ranging from SA benefiting grammatical accuracy (e.g., Arnett, 2013; Duperron, 2006; Guntermann, 1995; Howard, 2001, 2005; Isabelli, 2007; Isabelli & Nishida, 2005; Juan-Garau, Salazar-Noguera, & Prieto-Arranz, 2014; Linck, Kroll, & Sunderman, 2009; Mora & Valls-Ferrer, 2012; Rees & Klapper, 2007; Yager, 1998) to SA having no improvement (e.g., Arnett, 2013; Cheng & Mojica-Díaz, 2006; DeKeyser, 1991, 2010; Geeslin & Guijarro-Fuentes, 2005; Howard, 2001, 2005) to SA having detrimental effects (e.g., Collentine, 2004; Segalowitz, Freed, Collentine, Lafford, Lazar, & Díaz-Campos, 2004). Of interest to our study, off-line studies on morphosyntactic accuracy show no SA-NSA differences after four months of studying abroad (Isabelli-García, 2010: noun-adjective gender agreement with intermediate learners; Marqués-Pascual, 2011: SV agreement with intermediate and advanced learners).

Alongside this blurred scenario, online studies investigating SA effects on L2 morphosyntactic processing and WM effects in SA contexts are scarce. Online techniques are crucial to measure subtle changes in real-time processing efficiency that may not be obvious with off-line methodologies. For nominal agreement, Grey et al. (2015) examined the processing of noun-adjective nonadjacent grammatical gender and number agreement in advanced SA English learners of Spanish. The participants studied abroad for five weeks and completed a grammaticality judgment task, a lexical decision task, a reading span WM task (processing + storage), and a digit span short-term memory task (storage only). Relevant to our study, for the WM task,

participants listened to sets of L1 sentences and, for each sentence, memorized the last word (storage), judged whether the sentence made sense (Processing Task 1), and indicated whether it was grammatical (Processing Task 2). For the grammaticality judgment task, participants read L2 sentences and decided as quickly and accurately as possible whether each sentence was grammatical or not. The results indicated that SA increased accuracy on word order and nonadjacent noun-adjective number agreement, but not on gender agreement, and that WM was irrelevant.

Faretta-Stutenberg (2014) examined SA English-Spanish learners' processing of adjacent determiner-noun and noun-adjective grammatical gender agreement with event-related potentials (ERPs), a lower learner proficiency level (intermediate), a longer SA period (one semester), and an additional NSA group. Among other tasks, participants completed automated versions of the Operation Span, Reading Span, and Symmetry Span WM tests, and a grammaticality judgment task with rapid serial visual presentation that forced them to read sentences word by word at a fixed rate. The findings revealed that both groups evidenced some behavioral changes over the course of the semester, but only the SA group showed sensitivity to the determiner-noun agreement violations in the ERP results. Unexpectedly, there was a relationship between change in the response magnitude index and WM in the NSA, but not the SA, group. Interestingly, in a follow-up ERP study examining syntactic processing (noun-infinitive inversion) with the same participants and cognitive tasks, Faretta-Stutenberg and Morgan-Short (2017) found behavioral gains in both groups, processing gains only in the SA group, and WM effects on behavioral and processing gains in the SA, but not the NSA, group.

In a related study, Sagarra et al. (2016) employed eye-tracking and an implicit task (reading sentences and answering comprehension questions) to investigate how intermediate SA and NSA English-Spanish learners, and Spanish monolinguals process adjacent noun-adjective grammatical gender and number agreement. SA lasted one semester, and results revealed that all groups were sensitive to both violations: All looked longer at adjectives in number disagreement than agreement and at the word following the adjectives in gender disagreement than agreement. Also, only the monolinguals looked longer at adjectives in gender disagreement than agreement, but all regressed longer to adjectives in gender disagreement than agreement (longer regressions to number disagreement were unnecessary). This suggests that gender disagreement was more difficult than number disagreement. Importantly, NSA learners regressed more to the word following the adjective and were less sensitive to gender disagreement (the most difficult violation) than the other two groups. This indicates that the SA learners patterned closer to the monolinguals than their NSA counterparts.

Finally, LaBrozzi (2009, 2012) asked intermediate SA and NSA English-Spanish learners to complete a reading eye-tracking task with comprehension questions, a reading span WM test with plausibility judgments, and a Simon inhibitory control test. The target structure was adverb-verb and verb-adverb tense congruency (past adverbs with congruent past verbs or incongruent present verbs). The eye-tracking data showed that both groups were sensitive to tense incongruences, regardless of adverb position (before or after the verb): All looked longer at the word following the adverb in *verb-adverb than verb-adverb, and at the word following the verb in *adverb-verb than adverb-verb, and all regressed longer to the adverbs in *adverb-verb than adverb-verb. However, the SA learners patterned closer to Spanish

monolinguals than the NSA learners because the former relied more on morphological cues than the latter: only the SA group regressed longer to verbs in *verb-adverb than verb-adverb. Critically, WM effects applied to both groups for adverbs (higher WM span learners looked longer at adverbs in all conditions) but only to the SA group for verbs (higher WM span learners looked longer at verbs in *adverb-verb than adverb-verb). Finally, there were no inhibitory control effects for any of the groups.

Motivation for the current study

Most of the previous research employs off-line techniques and focuses on L2 grammatical accuracy. There are only four online studies on SA and L2 morphosyntactic processing. However, two employ grammaticality judgments (untimed: Grey et al., 2015; timed: Faretta-Stutenberg, 2014) and may not be measuring implicit knowledge. Of the two studies using a reading task with comprehension questions, Sagarra et al.'s (2016) study lacks a WM test, and LaBrozzi's (2009, 2012) study (i) combines morphosyntactic (adverb-verb tense congruency) and semantic processing (aspect) and (ii) counts PastAdverb-PresentVerb as incongruent when it could be acceptable in the historical present tense. Also, three of the aforementioned studies explore adjacent nominal agreement (Faretta-Stutenberg; Grey et al.; Sagarra et al., 2016) and two nonadjacent verbal agreement/congruency (Grey et al.; LaBrozzi). In addition, some studies focus on high-proficiency learners (Grey et al.), and others focus on low-proficiency learners (Faretta-Stutenberg; LaBrozzi; Sagarra et al., 2016). Finally, Grey et al. have a short-period SA (five weeks) and no NSA group. In sum, online studies about SA effects on L2 morphosyntactic processing show variability in tasks, linguistic structures, SA length, and presence of a NSA group. In the current study, we use eye-tracking methodology to investigate whether the SA benefits obtained in these studies extend to adjacent SV number agreement, which are easier to process than nonadjacent agreement, and implicit tasks.

In addition, we examine WM effects in NSA and SA low-proficiency learners. If WM is important for low-proficiency NSA learners, then it is critical for low-proficiency SA learners. This is because low-proficiency learners often rely on controlled processing, which requires time for planning and self-monitoring, a luxury that SA learners cannot afford. As a result, contextual pressures to keep the conversation flowing hinder learners' WM functions, thereby facilitating the automatization of incorrect L2 forms (Lafford, 2006). Thus, more advanced learners and those with high WM spans may benefit more from SA experiences because many grammatical structures have already been automatized. Conversely, learners with low WM spans or low proficiency will not benefit as much from SA due to their underdeveloped linguistic system and lack of automated processing. In addition, in SA contexts, the L1 is constantly inhibited. Thus, high-WM-span learners can benefit from SA because they can attend to multiple linguistic stimuli simultaneously (i.e., suppress the L1 and attend to the L2), whereas low-WM-span learners are unable to handle the crosslinguistic competition and either revert back to using the L1 or suffer other processing costs (Sunderman & Kroll, 2009). For example, in a single-word translation task, Tokowicz et al. (2004) found that low-WM-span learners and those with less SA experience made more nonresponse than meaning errors, whereas high-WM-span learners with more SA experience made as many nonresponse errors as meaning ones, because SA forces learners to attempt to make meaning instead of

not answering. In another lexical production task, Sunderman and Kroll (2009) reported that high-WM SA learners outperformed low-WM SA learners and high- and low-WM NSA learners, and concluded that low-WM-span learners cannot benefit as much from SA learning contexts. Online processing studies show WM effects with intermediate learners (LaBrozzi, 2009, 2012) but not with advanced learners (Grey et al., 2015) or with intermediate learners performing a taxing task (Faretta-Stutenberg, 2014). Following LaBrozzi, we consider whether the interaction of external factors (NSA vs. SA learning context) and internal factors (low vs. high WM span) shapes L2 morphosyntactic processing at the intermediate proficiency level.

Our research questions are as follows:

RQ1. Are NSA and SA intermediate English learners of Spanish sensitive to adjacent SV number agreement violations? If yes, are the SA learners superior to the NSA learners?

We hypothesize that both groups will be sensitive to the violations and that SA learners will be more sensitive than their NSA counterparts.

RQ2. Is WM capacity associated with intermediate NSA and SA learners' sensitivity to adjacent SV agreement violations?

As Faretta-Stutenberg and Morgan-Short (2017) and LaBrozzi (2009, 2012) found some WM effects with SA learners, but not with NSA learners, and because we are examining adjacent agreement, we only expect WM to facilitate sensitivity to the violations in the SA group.

Participants

Forty-seven native English learners of Spanish (25 SA, 22 NSA) participated in the study in exchange for extra credit and monetary compensation. All participants were right-handed (Oldfield, 1971), had normal or corrected-to-normal vision, began studying Spanish after the age of 12, had no knowledge of other foreign languages, had not lived in a non-English-speaking country for more than three months, completed all tasks, and were at least 75% accurate in the comprehension questions of the eye-tracking task. The original sample pool consisted of 50 participants, but 3 had to be excluded because they did not complete all of the tasks. Also, they were between 18 and 32 years old because processing speed and WM decline after the age of 40 (Park et al., 2003).

Before the program, all participants were sixth-semester Spanish students with a minimum GPA of 2.75 at the same large public American university, and the SA and NSA groups scored similarly in a Spanish proficiency test and in proficiency self-ratings (see the 'Results and Discussion' section). During the program, the SA group lived 16 weeks with a Spanish monolingual host family in a monolingual community in Spain and took 5–6 courses (15–19 credits) at the Institute for the International Education of Students (IES abroad) center in Madrid. The courses were taught in Spanish by teachers from the host institution and included art history, business, communications, economics, film studies, fine arts, geography, history, linguistics, literature, philosophy, political science, Spanish language, sociology, theater, and women's studies.

Materials and Procedure

Participants completed the tests individually in two sessions. First, they signed the consent form, performed an adapted version of the *Diploma de Español como Lengua Extranjera* (DELE) exam, and completed proficiency self-ratings (about 30 minutes). Shortly after the course/program, the two groups completed, in the following order, a language background questionnaire, a language contact profile (SA group), a WM test, an eye-tracking task, a vocabulary test, and a grammar test (about 90 minutes). Statistical analyses showed that the two groups were homogeneous in the WM, vocabulary, and grammar tests (see the ‘Results and Discussion’ section).

Screening Tests

The language background questionnaire was administered in English and contained questions about learners’ experience with Spanish before and after puberty, such as age of onset, number of years they studied Spanish, location and length of time living abroad, contact hours per week with Spanish, and context (in school, at home, with friends, at work). The language contact profile was from Freed, Dewey, Segalowitz, and Halter (2004), was given in English, and evaluated the quantity and quality of out-of-class L2 interaction during the SA sojourn. The adapted *DELE* proficiency test measured L2 grammatical knowledge via a 20-item, multiple-choice cloze test. The self-rating proficiency test assessed the four skills using a scale from 1 (minimum ability) to 5 (native-like proficiency). The vocabulary test consisted of a Spanish-English matching task with the target subjects and target verbs in infinitive. Finally, the grammar test was a Spanish-English matching task, with conjugated Spanish regular verbs in third-person singular and plural in the present and the preterit. The vocabulary and grammar tests were included to control for familiarity with the meaning of the target nouns and adjectives, and with the target structure (as a lack of familiarity with these items would increase RTs).

WM Test

This test was programmed with *E-Prime 2.0 Professional* (Psychology Software Tools) and was adapted from Waters and Caplan’s (1996) reading span test. Participants read 80 sentences in English silently at a fast pace, one by one, indicated whether each sentence was plausible by pressing a ‘yes’ or a ‘no’ button, and wrote down the final word of each sentence at the end of each set of sentences. Half of the sentences were plausible and half implausible with subject-object animacy inversion. Sentences were grouped into 20 sets of sentences and divided into 5 groups (span sizes 2–6 sentences) of 4 sets each. For accuracy scoring, participants received 1 point if their plausibility judgment was accurate, their recalled word was correct, and their response time was 300–5,000 ms long and 2.5 standard deviations above or below the mean. Because WM comprises *simultaneous* processing and storage, trials with a correct recall and an incorrect judgment, or correct judgment and incorrect recall were excluded. Response times faster than 300 ms and slower than 5,000 ms were omitted, because college students need 225–400 ms to process single words (Rayner & Pollatsek, 1989), and because unlimited sentence processing time would jeopardize the complexity of the test.

Eye-Tracking Task

The eye-tracker was an EyeLink 1000 (SR Research), with a sampling rate of 1 kHz, a spatial resolution of 0.32° horizontal and 0.25° vertical, and an averaged calibration error of 0.01°, and the experiment was programmed with *Experiment Builder* (SR Research). Participants read sentences in Spanish silently at their own pace and answered a yes-no comprehension question after each sentence. There were 146 sentences: 6 practice, 100 filler, and 40 experimental sentences. The filler and experimental sentences were divided into 10 blocks, and each block contained 4 experimental sentences (one per condition) and 10 filler sentences. Following a Latin square design, randomization occurred between and within blocks. All of the sentences were 9–15 words long. They included 2–4 syllable nouns and verbs that only appeared twice to reduce practice effects, and less than 20% of cognates to minimize lexical priming effects.

The experimental sentences followed a fixed syntactic order and contained animate human subjects and regular verbs in the third person of the present tense (indicative mood). The target subjects and verbs were selected from the Spanish textbook used in the participants' Spanish program. The experimental sentences had two conditions: SV agreement ($k = 20$: 10 S plural, 10 S singular) and SV disagreement ($k = 20$: 10 S plural, 10 S singular). For example, *Las enfermeras creen que los pacientes fuman/*fuma en el baño \ del pasillo* "The nurses think that the patients smoke/*smokes in the hallway's restroom" (\ indicates the break onto the next line). The comprehension questions were based on the content of the previously read sentence, instead of its grammaticality, so as not to draw participants' attention to SV agreement violations. For example, for the previous example sentence, the question was *¿El baño estaba en la habitación?* "Was the restroom in the room?" For all sentences, half of the comprehensions required a yes response and half a no response.

The eye-tracking task produced five scores: accuracy on the questions, gaze durations on S and V (time spent on the target word before moving on or looking back), and total time on S and V (duration of all fixations on the target word, including fixations from regressions). For accuracy, participants received 1 point for correct answers and 0 for incorrect ones.

Statistical analyses

Statistical analyses were divided into analyses comparing the NSA and SA groups to ensure the homogeneity of the group characteristics, and analyses of the eye-tracking and WM data. The first set of analyses included independent-samples *t*-tests (*SDs* are indicated in parentheses) and revealed that both groups were comparable in terms of WM, $t(45) = 0.579$, $p = 0.566$ (Levene = 0.228), NSA: $M = 45.09(13.40)$, SA: $M = 47.12(10.60)$; years learning Spanish, $t(24.14) = -0.165$, $p = 0.871$ (Levene = 0.000), NSA: $M = 6.95(0.95)$, SA: $M = 6.92(0.28)$; vocabulary test score (36 points), $t(45) = 0.090$, $p = 0.928$ (Levene = 0.857), NSA: $M = 35.91(0.43)$, SA: $M = 35.92(0.40)$; and grammar test score (32 points), $t(45) = 0.701$, $p = 0.487$ (Levene = 0.161), NSA: $M = 31.82(0.59)$, SA: $M = 31.92(0.40)$. In addition, both groups showed similar L2 proficiency on the *DELE* test (20 points), $t(45) = -0.929$, $p = 0.358$ (Levene = 0.348), NSA: $M = 10.27(1.86)$, SA: $M = 9.80(1.63)$; the reading self-ratings, $t(45) = 1.238$, $p = 0.222$ (Levene = 0.348), NSA: $M = 3.36(0.49)$, SA: $M = 3.56(0.58)$; the listening self-ratings, $t(45) = -0.751$, $p = 0.456$ (Levene = 0.213), NSA: $M = 3.45(0.74)$, SA: $M = 3.32(0.48)$;

Table 10.1 Descriptive statistics for WM, years learning

Measure	Group*	Low WM				High WM			
		Disagreement		Agreement		Disagreement		Agreement	
		M	SD	M	SD	M	SD	M	SD
Qu_Acc	NSA	76.19	12.23	81.82	09.24	83.98	07.46	85.71	09.28
	SA	79.76	14.09	79.37	10.22	85.35	07.64	87.55	07.15
Qu_RT	NSA	2132.67	396.70	2066.85	297.87	2017.30	465.44	1923.36	465.63
	SA	2185.77	567.87	2180.62	727.97	2057.56	290.60	1971.51	259.21
S Gaze	NSA	481.38	130.67	442.27	116.94	457.65	168.00	443.16	156.39
	SA	442.07	150.49	462.27	115.16	472.53	128.53	475.77	136.35
V Gaze	NSA	484.64	162.47	453.63	113.42	386.59	95.75	403.09	117.80
	SA	470.65	151.39	440.67	138.39	470.89	113.74	438.73	71.36
V+1 Gaze	NSA	236.66	38.81	246.75	64.06	220.27	37.90	241.13	35.27
	SA	259.48	81.98	247.78	41.11	243.13	53.50	248.18	48.44
S Total	NSA	771.07	301.46	723.85	239.81	720.23	277.21	689.32	259.72
	SA	676.80	265.31	691.10	221.20	740.49	189.92	703.30	158.80
V Total	NSA	705.36	245.95	670.89	229.45	579.30	200.94	557.84	191.98
	SA	625.22	269.12	583.89	265.29	641.93	200.60	560.53	90.83
V+1 Total	NSA	272.78	48.54	298.03	88.65	264.77	82.93	264.61	43.39
	SA	285.90	111.65	274.13	71.55	269.45	76.74	271.16	41.97

*NSA group: $n = 22$: 11 low WM, 11 high WM. SA group: $n = 25$: 12 low WM, 13 high WM.

the writing self-ratings, $t(45) = 1.715$, $p = 0.093$ (Levene = 0.418), NSA: $M = 3.32(0.72)$, SA: $M = 3.64(0.57)$; and the speaking self-ratings, $t(45) = 1.790$, $p = 0.080$ (Levene = 0.371), NSA: $M = 3.05(0.58)$, SA: $M = 3.32(0.48)$ (all self-ratings were out of 5 points).

The second set of analyses included four GLMMs with Agreement (Agree, Disagree), SA (SA, NSA), and WM (high, low), and all of their possible interactions as fixed factors, and Subject as a random factor. These GLMMs' dependent variables were accuracy and RTs on the comprehension questions, and RTs of gaze duration and total time on S, V, and V+1. Participants were divided into high or low WM using the median split. Accuracy data followed a binomial distribution (frequency logit link) and the rest a gamma distribution (frequency log link). Descriptive statistics are shown in Table 10.1.

Results and discussion

This study investigates the effects of learning context (SA, NSA) and WM on the on-line processing of L2 inflectional morphology in adult intermediate English learners of Spanish. Due to space limitations, only significant main effects and interactions are reported, and results and discussion are presented together.

RQ1: Sensitivity and SA Effects

The first research question explored whether the NSA and SA learners were sensitive to SV agreement violations and whether the SA learners were more sensitive than the

NSA learners. Our predictions that both groups would be sensitive and that the SA group would be more sensitive were supported. Thus, there was a significant main effect of Agreement, $F(1, 86) = 7.197, p = 0.009$, such that both groups were slower processing verbs (Total Time) in SV disagreement than agreement. These results follow online studies reporting that nonbeginning learners are sensitive to adjacent (e.g., intermediate: Sagarra & Herschensohn, 2010) and nonadjacent (e.g., intermediate: Havik et al., 2009; Sagarra & Ellis, 2013; advanced: Foote, 2011; Keating, 2010; Rossi et al., 2006) L2 morphosyntactic agreement violations.

Crucially, the interaction of Group $\times$ Agreement in processing verbs (Total Time), $F(1, 86) = 5.006, p = 0.032$, revealed that the SA group was more sensitive than the NSA group: SA: disagreement > agreement, $p = 0.009$; NSA: disagreement = agreement, $p = 0.253$. These findings indicate that the SA group was more sensitive to morphosyntactic agreement violations than the NSA group. This is in line with existing evidence showing the superiority of SA over NSA learners on adjacent (Sagarra et al., 2016; Faretta-Stutenberg, 2014 with behavioral data) and nonadjacent (LaBrozzi, 2009, 2012) L2 morphosyntactic agreement violations. Grey et al. (2015) lacked an NSA group, but their SA group improved from the pretest to the posttest in noun-adjective number agreement. The absence of improvement in noun-adjective gender agreement can be explained by the numerous studies showing that gender agreement is cognitively more demanding than number agreement (Antón Méndez, Nicol, & Garrett, 2002; Gillon Dowens, Barber, Vergara, & Carreiras, 2010; Rossi, Kroll, & Dussias, 2014; Sagarra & Herschensohn, 2010). Because gender agreement is difficult to acquire and the SA only lasted five weeks, the lack of SA benefits is not only justified but also expected. In sum, the findings reveal that the NSA and SA learners were sensitive to adjacent SV number agreement violations and that the SA learners were more sensitive than the NSA learners.

RQ2: WM Effects

The second research question examined whether WM was associated with sensitivity to SV agreement violations in NSA and SA learners. Our hypothesis that WM would affect SA learners', but not NSA learners', sensitivity to the violations was supported by pairwise comparisons. Although there was not a significant effect of WM in processing V (Total Time), the pairwise comparisons of the interactions of WM $\times$ Agreement, $F(1, 86) = 0.165, p = 0.685$, and of WM $\times$ Agreement $\times$ Group, $F(1, 86) = 0.635, p = 0.428$, revealed promising findings. The two-way interaction showed that high-WM-span learners were sensitive to the violations: high -WM-span learners showed longer RTs in sentences with SV disagreement than agreement, $p = 0.030$, whereas low-WM-span learners showed similar RTs in sentences with SV disagreement and agreement, $p = 0.114$. Importantly, this WM effect applied to an implicit task. These findings are in line with studies reporting WM effects on L2 grammar development (Havik et al., 2009; Sanz et al., 2016) and L2 morphosyntactic processing (LaBrozzi, 2009) with implicit tasks, and demonstrate that WM effects occur independently of the nature of the task (Engle, 2002; Reber, Walkenfeld, & Hernstadt, 1991; Roberts, 2012, among others, propose that WM effects only occur with explicit tasks). Critically, the three-way interaction indicated that, within the high-WM-span learners, only the SA group was sensitive to the violations (disagreement > agreement, $p = 0.012$, compared to disagreement = agreement, $p = 0.212$ in SA

low-WM-span learners, $p = 0.538$ in NSA high-WM-span learners, and $p = 0.318$ in NSA low-WM-span learners). Because the aforementioned two- and three-way interactions were not significant, we should interpret the results of the pairwise comparisons with caution. However, the lack of significance can easily be explained by the trend for disagreement to be higher than agreement applied to both groups. Thus, even though these interactions were not significant, their pairwise comparisons clearly suggest the superiority of SA learners with higher WM span over the rest.

The pairwise comparison findings of the two-way interaction are in line with studies reporting WM effects on L2 morphosyntactic processing (Faretta-Stutenberg, 2014; Sagarra, 2008; Sagarra & Herschensohn, 2010; Sagarra et al., 2016) and L2 grammar development (Linck & Weiss, 2011, 2015; Sanz et al., 2016; Serafini & Sanz, 2016, with lower proficiency learners). In turn, the pairwise comparison results of the three-way interaction are in agreement with studies showing WM effects with SA learners regarding L2 morphosyntactic processing (LaBrozzi, 2009) and lexical comprehension and production (Sunderman & Kroll, 2009; Tokowicz et al., 2004). Noteworthy, Grey et al. (2015) found no WM effects on SA learners' L2 morphosyntactic processing, but there was no NSA comparison group. Moreover, gender agreement is late-acquired, and their adapted version of the reading span WM test was more difficult than the standard reading span test: two processing tasks (plausibility judgment, grammaticality judgment) instead of one (plausibility judgment). In turn, Faretta-Stutenberg (2014) found WM effects with NSA, but not SA, learners; however, the two groups were not compared, and she reported different findings (viz. WM effects with SA, but not NSA, learners) in a 2017 follow-up study with Morgan-Short using the same participants.

In addition to the RT results discussed earlier, there was a significant main effect of WM, $F(1, 86) = 7.393$, $p = 0.008$, in accuracy on the comprehension questions, such that both groups were more accurate answering the questions if they had higher than lower WM span. These findings indicate that WM facilitated semantic processing at intermediate proficiency levels, independently of SA experience. The association between WM and L2 reading comprehension has been widely documented in beginning and intermediate learners, both in cross-sectional studies (e.g., Abu-Rabia, 2003; Harrington & Sawyer, 1992; Leiser, 2007; Walter, 2004) and in longitudinal studies (e.g., Kormos & Safar, 2008; Linck et al., 2012; Sagarra & LaBrozzi, forthcoming). To conclude, the results suggest that higher WM capacity facilitates (i) L2 morphosyntactic processing in SA, but not in NSA, learners, and (ii) semantic processing in both SA and NSA learners.

Implications: Recommendations for Practice

The number of university students studying abroad is on the rise. However, SA programs vary widely, and most research on SA effects on SLA employs off-line measures and focuses on oral fluency, pronunciation, sociolinguistic competence, and cultural awareness. This study provides strong evidence for the positive impact of SA on L2 morphosyntactic processing and development. We propose that SA's abundance of nonredundant morphological cues and of opportunities for hypothesis testing facilitates attention to inflectional morphology, and we recommend NSA classroom practices to implement structured input activities lacking lexical cues. Our findings also suggest that WM modulates L2 morphosyntactic processing in SA learners, an

indication that processing of inflectional morphology is cognitively taxing. We advise language instructors to implement pedagogical practices that facilitate L2 processing by reducing attentional demands and cognitive load (e.g., via input simplification and simple tasks) and making suffixes salient (e.g., via input enhancement). Finally, our results suggest that learning context and WM play a key role during sentence processing and should be part of SLA theories (Collentine & Freed, 2004).

Limitations/Future Directions

This study provides strong evidence that a 16-week SA sojourn facilitates adult intermediate learners' sensitivity to L2 morphosyntactic agreement violations during visual sentence comprehension. Future research with nonadjacent violations, oral stimuli, and different SA program durations, linguistic structures, proficiency levels, and target languages will broaden our understanding of SA benefits and help us to elucidate what it is about this experience that accounts for divergent processing strategies between SA and NSA learners. We also found robust WM effects on L2 semantic processing (accuracy on the questions) and weak effects on L2 morphosyntactic processing (RTs). This imbalance could be due to the reading nature of the reading span test (testing reading through reading) and the WM scoring method (recall accuracy). The implementation of nonlinguistic WM tests and additional statistical analyses with processing speed scores (e.g., plausibility judgment RTs) may generate different outcomes.

Key Terms

Study abroad	(Cognitive) Individual differences
L2 processing	L2 development
Eye-tracking	Morphosyntactic processing
Working memory	Semantic processing

Further Reading

- DeKeyser, R. M. (2014). Research on language development during study abroad. In C. Pérez-Vidal (Ed.), *Language acquisition in study abroad and formal instruction contexts* (pp. 313–326). Philadelphia, PA/Amsterdam, The Netherlands: John Benjamins. (Provides a brief overview of SA research and suggestions for future SA research. It addresses methodological issues, such as the marrying of quantitative and qualitative research.)
- Grey, S., Cox, J. G., Serafini, E. J., & Sanz, C. (2015). The role of individual differences in the study abroad context: Cognitive capacity and language development during short-term intensive language exposure. *The Modern Language Journal*, 99(1), 137–157. (Considers the roles of L2 proficiency, study abroad duration, and cognitive individual differences in morphosyntactic and lexical development via a grammaticality judgment task and lexical decision task, and reveals that individual differences were not related to L2 development but that advanced students spending five weeks abroad does lead to significant L2 gains.)
- Yang, J. (2016). The effectiveness of study abroad on second language learning: A meta-analysis. *The Canadian Modern Language Review*, 72(1), 66–94. (Reviews previous findings via a meta-analysis of methodological features and quantitative studies on L2 development in SA and NSA contexts, and reveals that SA learners perform better than NSA learners and also that a short-term experience abroad is more advantageous to SLA than a long-term one.)

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